Volumetric Physical Data Storage

Color, structure, depth, growth, and scan-readable patterns

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Abstract

This paper proposes a feasibility study of data encoded at registered positions within physical objects. Candidate paths combine optical response with written structure, recover internal patterns by tomography, or encode data during crystal growth or fabrication. Depth supplies additional storage locations. Growth-defined properties add capacity when separately controllable, readable states expand the joint alphabet. Calibration, error correction, and independent readback determine usable density and system energy. Existing optical storage, dopant-modulated growth, and embedded codes establish substantial prior work. The immediate goal is a small reproducible recovery test. No physical prototype, demonstrated 32-bit voxel, fabrication-encoded archive, or verified capacity, durability, or energy advantage is reported.

1. Origin and research question

The author reports an earlier experiment converting digital data into colors and printing the result as a physical display. Recovering the data proved difficult when images were resized or file handling changed the representation. No original samples, measurements, or reproducible decoder from that experiment were evaluated for this paper. It supplies the motivation, not performance evidence.

The proposed next question is: can a stable transparent medium, combined with a calibrated reader and a self-describing format, recover multistate optical data reliably enough to justify its physical density and energy costs? Glass is a candidate; a crystalline or engineered host would require its own measurements of transparency, write response, and stability.

2. Relationship to existing research

Microsoft Research reported an end-to-end glass-storage system in 2026, including 4.8 TB in a 120 mm square, 2 mm thick plate with 301 layers. Its encoded optical structures and error-correction pipeline demonstrate the broader feasibility of volumetric glass archives [1]. Southampton has demonstrated laser-written storage based on spatial position and optical properties of internal nanostructures [2].

Thus, storing data inside glass and using multiple optical states are established research directions. The proposed contribution here is a test plan for calibrated color or spectral states alongside other optical measurements. Whether this combination offers anything new or better must be established through a fuller literature review and controlled comparison.

3. Proposed encoding and recovery

A localized storage element is called a voxel. Its x, y, and z coordinates identify its location. Its reproducibly measurable optical response identifies its symbol. Candidate responses include spectral absorption or transmission, polarization response, and structural properties. These channels must be assessed for independence: two measurements of the same physical change do not necessarily carry two independent sets of bits.

Simple surface etching does not establish the ability to create arbitrary permanent colors inside glass. Candidate processes could include laser-induced structural changes or suitably modified glass compositions. The appropriate process, host material, and read wavelengths are open experimental questions. Strong absorption or scattering could obstruct deeper layers, reducing the advantage of visible color.

Write path: source bytes; optional lossless compression; format header and block identifiers; error-correction encoding; mapping to qualified optical symbols; writing into registered voxel positions; independent readback verification.

Read path: locate reference marks; estimate grid and depth; calibrate against known optical references; measure voxel responses; classify symbols and flag uncertainty; decode error correction; verify block and file hashes; reconstruct exact source bytes.

Each block should carry a format version, dimensions, symbol mapping, payload length, and recovery parameters. Reference marks should permit geometric registration. Calibration marks should span the measured optical response. Redundancy should be distributed to reduce vulnerability to a single scratch, fracture, or unreadable layer. A hash detects an incorrect reconstruction; it does not itself repair the data.

4. Capacity: conditional arithmetic

For M distinguishable states, the maximum fixed-length symbol size is floor(log2 M) bits. Thus, 256 qualified states can encode 8 bits; 2^32 qualified states can encode 32 bits. Computer-style 32-bit color commonly includes transparency and does not prove that a physical dot has 2^32 readable states.

For lateral pitches px and py, layer pitch pz, symbol size b, and usable fraction f, approximate usable density is b f / (8 px py pz) bytes per unit volume. The fraction f includes coding and reserved-space losses; margins and packaging must also be counted. This arithmetic assumes the optical system can resolve every voxel at those pitches.

Conditional design for 1 TB	8 bits/voxel	32 bits/voxel
Single layer; 1 micrometer lateral pitch	100 x 100 cm	50 x 50 cm
1,000 layers; same lateral pitch	3.16 x 3.16 cm	1.58 x 1.58 cm
Approximate depth; 10 micrometer layer pitch	1 cm	1 cm

1 TB = 10^12 bytes. Values exclude overhead and are geometric examples, not demonstrated designs. Depth is used here for separate layers; it must not also be counted as extra symbol bits for those same occupied positions.

5. Why reading reliability governs capacity

A large palette helps only if neighboring states remain separable after writing variation, illumination changes, sensor noise, focus errors, and interference from other voxels. Increasing the number of nominal states can increase error rates and recovery overhead enough to reduce useful capacity.

An illustrative calculation places M values uniformly along one normalized scalar measurement. Assume zero-mean Gaussian measurement noise with standard deviation 0.001 of full scale. Interior-symbol error probability is approximately 2 Q[1 / (2 (M - 1) sigma)], where Q is the standard normal upper-tail probability. At M = 256 this is approximately 5%; at M = 16 it is negligible under these assumptions. These are analytical model results, not measured glass errors. Endpoints have different error probabilities.

The calculation does not model RGB color, wavelength-resolved measurements, multidimensional palettes, material physics, or correlated errors. In particular, it cannot decide whether a real 256-color system is feasible. A measured multidimensional confusion matrix is needed. The 32-bit target should remain a speculative upper target until actual channel capacity and reliability are established.

6. Small experimental program

Stage A - decoder and printed control. Encode known binary payloads using 4, 16, and 256 symbols, reference marks, and calibration patches. Test optical capture under rotation, scale changes, different illumination, and controlled blur. Compare exact recovered bytes, raw symbol errors, coding overhead, and failed blocks. Pixel merging or removal must be reported as information loss, not treated as automatically reversible resizing.

Stage B - single-layer material sample. Select a writing process with an optical-materials laboratory. Measure stable achievable states before assigning a palette. Write repeated randomized samples at several pitches. Use separately written calibration and evaluation samples; report the confusion matrix and uncertainty. Compare a simple two-state design against each multistate design using the same physical area.

Stage C - depth experiment. Begin with a few separated layers. Vary depth and spacing, then measure focus tolerance, signal attenuation, scattering, and errors caused by other layers. Increase layer count only while verified usable capacity improves. Avoid extrapolating a small shallow sample directly to 1,000 layers.

Stage D - recovery and durability. Recover previously unseen payloads with the decoder fixed in advance. Test repeat reads, temperature changes, and selected damage scenarios. Document failed recovery as well as successes. Repeat across multiple samples and write sessions. Accelerated ageing requires a justified model before any lifetime claim.

7. Decision criteria

Advance the design only if exact payload recovery is reproducible and additional states or layers increase usable bytes per volume after overhead. A proof of concept should demonstrate full-file hash agreement across independently written samples and all declared operating conditions. The team must specify sample size, raw error limits, acceptable unrecoverable-block rate, and confidence bounds before testing. Zero failures in a small trial is not proof of archival reliability.

8. Energy, space, and intended use

A passive glass object requires no electrical power simply to retain its marks. That does not make its storage system energy-free, and passive retention alone is not an advantage over every other offline medium. Lasers, motion systems, sensors, computers, manufacturing, and retrieval equipment all contribute costs. Long-term archives are the proposed application; frequent modification would require a separate update strategy.

A fair comparison must define an archive duration and access schedule and measure energy for writing, verifying, reading, handling, and maintaining the system. Include replacement and migration assumptions, manufacturing where measurable, spare copies, and failed writes. Compare usable bytes per packaged volume and energy per verified recovered byte against suitable offline alternatives. Reduced physical volume and reduced energy are separate outcomes; neither follows automatically from a higher symbol count.

9. Collaboration request

The author seeks an initial technical assessment of achievable optical states, suitable host materials, available writing and reading equipment, and a small experiment. The immediate request is to identify a feasible test and its limitations, rather than endorse a terabyte-scale design. A student project on decoding and calibration could precede a materials experiment. Specialized laser work should be planned and performed by a qualified laboratory.

Questions for a research partner: Which stable optical channels can be written and read in the same voxel? How many distinguishable states survive repeated independent measurements? At what pitches does cross-talk erase the benefit? Does a multistate design outperform a simpler design after correction overhead? What is its whole-system energy for a defined archive workload?

10. Status and scope

This is an independent concept paper for discussion. No university affiliation, endorsement, physical prototype, performance guarantee, or claim of first invention is asserted. Capacity examples are conditional calculations. The earlier printed-color experiment is author-reported. The reading-noise example is a simplified analytical model. The references establish relevant prior work, not validation of this proposed implementation.

References

[1] Microsoft Research Project Silica Team. "Laser writing in glass for dense, fast and efficient archival data storage." Nature 650, 606-612 (2026). Published February 18, 2026. DOI: 10.1038/s41586-025-10042-w. https://www.nature.com/articles/s41586-025-10042-w

[2] University of Southampton. "Eternal 5D data storage could record the history of humankind." February 2016. Institutional explanation of laser-written nanostructures and polarization-based readout. https://www.southampton.ac.uk/news/2016/02/5d-data-storage-update.page

Sources accessed October 11, 2026. This brief is not an exhaustive literature or patent review.

Appendix A. Plastics, synthetic crystals, and gemstones

The concept can be studied in transparent hosts beyond glass. Optical clarity is necessary for many depth-reading schemes, but it is insufficient: the material must also support stable, distinguishable written states and repeated recovery. Polishing improves the external surface; it does not remove internal inclusions, absorption, strain, or crystallographic optical effects. A flat optical plate is a more controllable starting geometry than a faceted gemstone.

Engineered polymers. Specialized light-responsive polymers offer a plausible route to three-dimensional optical recording. A 2024 Nature paper reports volumetric nanoscale optical storage in a dye-doped photoresist [3]. This does not establish that ordinary clear acrylic or packaging plastic will store the proposed symbols. Candidate samples should be tested for dimensional change, optical ageing, and loss of readable contrast under the intended conditions.

Synthetic sapphire. Sapphire is a candidate transparent crystalline host. Researchers have demonstrated laser-inscribed birefringent structures in sapphire [4], which establishes a relevant material response, not an archival storage system. Crystal orientation, native birefringence, scattering from modifications, and writing stress would need evaluation before assigning data states. A polished synthetic sample could be screened against glass under the same decoder and measurement protocol.

Diamond. Experiments have optically patterned trapped charge in nitrogen-doped diamond [5]. This is a different storage mechanism from ordinary colored etched dots. It supports investigating engineered defects in a gemstone material, but does not validate 32 bits per voxel, unrestricted readout, or long-term retention for this proposal. Synthetic diamond should be considered only if a research partner identifies a practical mechanism and suitable equipment.

Natural gemstones and synthetic quartz. Natural stones would require individual characterization of internal defects and optical uniformity. Synthetic crystals offer an opportunity to specify composition and growth conditions, but are not automatically better storage media. Crystalline quartz and fused silica glass are distinct hosts; performance established in one must not be transferred to the other without testing.

Proposed screening order. Begin with a host and process a partner laboratory can already write and read. Compare an engineered polymer or glass control with one promising synthetic crystal. Rank candidates by verified bytes per packaged volume, energy per recovered byte, stability, repeatability, and equipment cost. Gemstone appearance or hardness is not a substitute for these measurements.

Additional references

[3] Zhao et al. "A 3D nanoscale optical disk memory with petabit capacity." Nature 626, 772-778 (2024). DOI: 10.1038/s41586-023-06980-y.
https://www.nature.com/articles/s41586-023-06980-y

[4] "Laser-Inscribed Stress-Induced Birefringence of Sapphire." Nanomaterials 9(10), 1414 (2019). DOI: 10.3390/nano9101414.
https://www.mdpi.com/2079-4991/9/10/1414

[5] "Optical patterning of trapped charge in nitrogen-doped diamond." Nature Communications 7, 12660 (2016). DOI: 10.1038/ncomms12660.
https://www.nature.com/articles/ncomms12660

Appendix B. Combined properties and scan-readable objects

Color plus structure. A voxel could carry a color or spectral value together with a shape or orientation value. If 16 optical states and four orientations can each be reliably distinguished in every combination, the joint alphabet has 64 states and carries six bits. With 256 optical states and four orientations, it has 1,024 states and carries ten bits. These are conditional examples. Etching that changes the color response or forces larger spacing can erase the gain. A full joint confusion matrix is required before capacities are multiplied.

Researchers have combined fluorescence intensity with laser-written structure orientation in silver-containing glass [6]. This is relevant prior work, not validation of arbitrary visible colors. The question is whether an added channel improves recovered density after overhead and errors.

Printed internal patterns. Encode bytes as cavities, solid regions, or composition changes inside a printed host. X-ray tomography reconstructs the interior. Embedded authentication codes have been recovered from manufactured parts using micro-CT [7]; general archival storage still requires capacity, reliability, and energy measurements.

MRI requires suitable magnetic-resonance properties. CT is a more direct candidate for cavities. Visible colors need not differ under X-rays; spectral CT distinguishes selected compositions [8], but does not establish a 256-material palette. Combined optical and CT readout would require access to both signals and precise registration.

Conditional scale. A 10 cm cube with one bit per cubic cell would hold 125 KB at 1 mm cell pitch, 125 MB at 0.1 mm, or 125 GB at 0.01 mm, before overhead. These are geometric counts. Cell pitch must accommodate actual walls, write accuracy, and adequate imaging resolution; a reconstructed scan pixel does not automatically correspond to an independent recoverable bit. A large brain-scan file is not evidence of equivalent object storage capacity.

First test and protection. Print a small block containing a known payload, orientation markers, calibration regions, correction data, and a hash. Recover the bytes from scans at several cell sizes and object orientations. Compare against a matched single-property control. Internal placement may protect against surface wear, but cracks, attenuation, reader obsolescence, and missing recovery instructions remain concerns. It provides neither encryption nor a demonstrated net energy saving.

Additional references

[6] Park et al. "Five-Dimensional Optical Data Storage Based on Ellipse Orientation and Fluorescence Intensity in a Silver-Sensitized Commercial Glass." Micromachines 11(12), 1026 (2020). DOI: 10.3390/mi11121026. https://www.mdpi.com/2072-666X/11/12/1026

[7] "Embedded product authentication codes in additive manufactured parts: Imaging and image processing for improved scan ability." Additive Manufacturing 35, 101319 (2020). DOI: 10.1016/j.addma.2020.101319. https://pmc.ncbi.nlm.nih.gov/articles/PMC8017490/

[8] "Multicolor spectral photon-counting computed tomography: in vivo dual contrast imaging with a high count rate scanner." Scientific Reports (2017). DOI: 10.1038/s41598-017-04659-9. https://www.nature.com/articles/s41598-017-04659-9

Appendix C. Encoding data during crystal growth

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Research question. Can a digital payload be mapped to controlled changes during crystal growth, then recovered exactly from the finished object? A further question is whether the resulting material pattern can carry data alongside color or spectral response and later-written structure. These are proposed experiments; this paper reports no fabrication-encoded storage prototype.

Relevant prior work. Kurlov and Belenko grew sapphire with spatially modulated dopant concentrations [9]. Lobaev et al. controlled nitrogen incorporation during chemical vapor deposition (CVD) diamond growth and reported thin nitrogen-doped layers [10]. These results establish material control, rather than exact-file archival recovery. Combined fluorescence intensity and written-mark orientation in glass [6] provide a separate precedent for using more than one readable property.

Candidate process. First qualify two reproducibly distinguishable material responses. A growth recipe would switch between them to encode a sequence of bands or regions, with reference marks, calibration, and correction data. Subsequent growth could cover the encoded region; its effect on contrast and recovery must be measured. Arbitrary local control across a growing layer is a further requirement for dense three-dimensional patterns. A grown-in material pattern and a pattern in a printed composite require different fabrication and validation methods.

Combining readable properties. The following alphabet is a conditional example. Each state count must be qualified experimentally, and every intended combination must remain selectable and distinguishable at the same independently controlled storage location.

Proposed property	Qualified choices assumed
Color or spectral response	16
Written structure or orientation	4
Separate material state set during growth	2

The full 16 x 4 x 2 alphabet would contain 128 combinations, or seven bits per location, before overhead. The 16 x 4 control would contain six bits. A measured joint confusion matrix must establish recoverability. If growth creates the color being measured, that encoded value is counted once. Extra properties can require larger spacing or more correction, reducing net gain.

Spatial granularity. If N six-bit optical locations share one two-state growth band, their ideal total is 6N + 1 bits, assuming the band value is also recoverable. It is not 7N bits: the shared growth value has only one independently selectable choice for the whole band. Depth used to address separate locations must likewise be counted once in the physical layout.

Appendix C (continued). Experiment and decision criteria

First demonstration. Begin with a short payload, such as a 16-bit message, fabricated as a sequence of distinguishable growth regions. Record the format, synchronization marks, calibration regions, correction parameters, and verification hash. The short message tests the process; its small sample size cannot establish archival reliability.

1. Qualify the material states. A partner laboratory should select a host and process it can already grow and measure. Separate calibration and evaluation samples. Measure state contrast, boundary width, variation across sessions, readout uncertainty, and the effective spatial granularity. A thin grown layer is not automatically a separately readable optical symbol.

2. Fabricate and recover unseen payloads. Fix the decoder and symbol mapping before independently preparing randomized evaluation samples. Withhold source payloads from the reader. Recover exact bytes and compare their hashes; record raw errors, corrected errors, failed blocks, and manufacturing failures. Include tests before and after any covering growth.

3. Test the combined design. Compare growth-only samples, color-plus-structure controls, and the proposed combined samples. Exercise all intended joint states, rather than a few favorable examples. Check whether later writing changes the grown code, whether growth states alter the other measurements, and whether these interactions increase required spacing.

4. Compare the whole process. Report verified payload bytes per packaged volume after all reference, format, coding, and protective-material overhead. Compare growth time, subsequent writing time, verification, readout, manufacturing yield, and measured energy for matched workloads. Combining manufacture and writing may eliminate an operation, but slower growth or harder readout can offset that benefit.

Decision gate. Specify sample counts, operating conditions, error limits, and confidence reporting before testing. Advance only when exact recovery is reproducible and the additional property gives a measured capacity or manufacturing benefit under a defined comparison. Assess attenuation, growth-induced stress, boundary smearing, and reader access as material-specific risks. No novelty, lifetime, cost, or energy advantage is established here.

Revision record. Version 1.2 updates the cover, abstract, and capacity-table typography and adds Appendix C. The original discussion and numerical examples are retained. Version 1.1 remains available in the repository for reference.

Additional references

[9] V. N. Kurlov and S. V. Belenko. "Growth of sapphire shaped crystals with continuously modulated dopants." Journal of Crystal Growth 191(4), 779-782 (1998). DOI: 10.1016/S0022-0248(98)00396-0. https://doi.org/10.1016/S0022-0248(98)00396-0

[10] M. A. Lobaev et al. "Influence of CVD diamond growth conditions on nitrogen incorporation." Diamond and Related Materials 72, 1-6 (2017). DOI: 10.1016/j.diamond.2016.12.011. https://doi.org/10.1016/j.diamond.2016.12.011

References [9]-[10] checked October 12, 2026. Reference [6] is listed in Appendix B. This addition is a feasibility proposal, not an exhaustive literature review.